

Unitizations of Separable-Fibre Continuous-Trace Algebras Have BCP

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when its pure state space, i.e. K , has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

The previous continuous-trace packet proves the expected criterion for continuous-trace algebras whose elementary fibres are separable. The present packet proves that the positive theorem survives one scalar multiplier direction.

Theorem 1 (unitized continuous-trace subcase). *Let I be a continuous-trace C^* -algebra whose elementary fibres are $\mathcal{K}(H_x)$ with H_x separable and nonzero. Let $A = \tilde{I}$ be the minimal unitization of I when I is nonunital; if I is unital, put $A = I$. Then*

$$A \text{ has BCP} \iff A \text{ has UBCP} \iff \text{Prim}(I) \text{ has a countable } \pi\text{-basis.}$$

Idea of the Proof

The old compact-bump proof works because local compact-operator sections can be multiplied by scalar bumps. In a unitization, an element has local form

$$a(x) = \lambda I + i(x),$$

and λI cannot be localized when the fibre is infinite-dimensional. This is the new obstacle.

The trick is to use the scalar direction as margin rather than trying to localize it. If $|\lambda|$ is tiny, the compact part $i(x)$ nearly norms the element and the old compact proof still works. If $|\lambda|$ is bounded below, choose a finite-rank approximation T to the local compact part and add a large rank-one compact “ballast” $M\omega Q$ in a direction orthogonal to T . The center is

$$c = \mu 1 + h(T + M\omega Q),$$

where μ approximates λ , ω has the same phase, and h is a local bump. On the spike direction, $\|c\|$ is about $M + |\lambda|$, while the distance from a to c is only about M . That fixed scalar margin is exactly what the naive localization argument was missing.

Proof

We use two standard facts already used in the companion packets.

Lemma 1 (primitive π -base obstruction). *If a C^* -algebra B has BCP, then $\text{Prim}(B)$ has a countable π -basis.*

Lemma 2 (charted countable bump families). *Assume that $X = \text{Prim}(I)$ has a countable π -basis. There are countably many continuous-trace trivialization charts*

$$\Phi_n : I_{O_n} \longrightarrow C_0(O_n, \mathcal{K}(H_n)),$$

where H_n is separable and nonzero, such that (O_n) is a π -basis of X . For each n , there is a countable family $\mathcal{B}_n \subset C_c(O_n)$ with $0 \leq h \leq 1$, $\|h\|_\infty = 1$, and such that every nonempty open $W \subset O_n$ contains the support of some $h \in \mathcal{B}_n$.

Proof. This is the chart-and-bump selection from the continuous-trace packet. Local triviality of continuous-trace algebras gives charted neighbourhoods. Refining the given countable π -basis by these charted neighbourhoods gives a countable chart π -basis. Since each O_n is locally compact Hausdorff, compactly supported scalar bumps with cofinal supports are obtained from local regularity. $\square$

Proof of Theorem 1. If I is unital, then $A = I$, and the result is exactly the previous continuous-trace separable-fibre theorem. We therefore assume that I is nonunital and write elements of A uniquely as

$$a = \lambda 1 + i, \quad \lambda \in \mathbb{C}, \quad i \in I.$$

The implication $\text{UBCP} \Rightarrow \text{BCP}$ is immediate. If A has BCP, Lemma 1 gives a countable π -basis for $\text{Prim}(A)$. The open subset $\text{Prim}(I) \subset \text{Prim}(A)$ then also has a countable π -basis. It remains to prove UBCP assuming that $\text{Prim}(I)$ has a countable π -basis.

Fix the charts and bump families from Lemma 2. We shall build countably many balls covering S_A and uniformly avoiding a neighbourhood of the origin.

First include the old compact-bump centers. For each n , choose a countable $1/64$ -net $D_n \subset S_{\mathcal{K}(H_n)}$. For $h \in \mathcal{B}_n$ and $T \in D_n$, include

$$c_0(n, h, T) = \Phi_n^{-1}(4hT) \in I \subset A$$

with ball radius $7/2$. These centers have norm 4.

Next include scalar-ballast centers for the infinite-dimensional charts. Put

$$\eta = \frac{1}{16}, \quad \varepsilon = \frac{1}{256}, \quad M = 4.$$

For each infinite-dimensional H_n , fix an orthonormal basis $(e_m)_{m \geq 1}$. Let $\mathcal{F}_n$ be the countable set of finite-rank operators whose matrices, relative to this basis, have rational complex entries and finite support. This set is norm dense in $\mathcal{K}(H_n)$. Let $\mathcal{Q}_n = \{Q_m : m \geq 1\}$, where Q_m is the rank-one projection onto $\mathbb{C}e_m$. Let $\Lambda \subset \mathbb{C}$ be a countable dense set and let $\Omega \subset \mathbb{T}$ be a countable dense set. For every

$$n, h \in \mathcal{B}_n, T \in \mathcal{F}_n, Q \in \mathcal{Q}_n, \mu \in \Lambda, \omega \in \Omega$$

include the center

$$c_1(n, h, T, Q, \mu, \omega) = \mu 1 + \Phi_n^{-1}(h(T + M\omega Q)).$$

We retain in the covering family only those of these countably many centers which satisfy

$$\|c_1\| \geq M + \eta - 2\varepsilon,$$

and for them use the fixed radius $M + 3\varepsilon$. Since

$$M + \eta - 2\varepsilon - (M + 3\varepsilon) = \eta - 5\varepsilon > 0,$$

these balls are uniformly off the origin.

We now prove that the two families cover S_A . Let $a = \lambda 1 + i \in S_A$. For $P \in O_n$, write

$$i_n(P) = \Phi_n(i|_{I_{O_n}})(P) \in \mathcal{K}(H_n).$$

Suppose first that either $|\lambda| < \eta$, or the selected chart below has finite-dimensional fibre. The set

$$W = \{P \in \text{Prim}(I) : \|\lambda I + i(P)\| > 3/4\}$$

is nonempty and open; if the norm of a is attained only at the quotient character, then $|\lambda| = 1$, so this case cannot occur. Choose a chart $O_n \subset W$ and $P_0 \in O_n$.

If H_n is finite dimensional, set

$$b(P) = \lambda I_{H_n} + i_n(P) \in \mathcal{K}(H_n).$$

If H_n is infinite dimensional, set $b(P) = i_n(P)$. In the infinite case,

$$\|b(P_0)\| \geq \|\lambda I + i_n(P_0)\| - |\lambda| > \frac{3}{4} - \eta > \frac{11}{16}.$$

In the finite-dimensional case, $\|b(P_0)\| > 3/4$. Thus in both cases $\alpha = \|b(P_0)\| > 11/16$. Choose $T \in D_n$ with

$$\left\| T - \frac{b(P_0)}{\alpha} \right\| < \frac{1}{64}.$$

For $0 \leq s \leq 1$, convexity gives

$$\|b(P_0) - 4sT\| \leq \max\{\alpha, 4 - \alpha(1 - 1/64)\} < \frac{7}{2} - \eta.$$

By norm-continuity of the compact section b , there is a nonempty open neighbourhood $N \subset O_n$ of P_0 such that

$$\sup_{0 \leq s \leq 1} \|b(P) - 4sT\| < \frac{7}{2} - \frac{\eta}{2} \quad (P \in N).$$

Choose $h \in \mathcal{B}_n$ with $\text{supp } h \subset N$, and put $c = c_0(n, h, T)$. If H_n is finite dimensional, then on $\text{supp } h$

$$\|a(P) - c(P)\| < 7/2.$$

If H_n is infinite dimensional, then on $\text{supp } h$

$$\|a(P) - c(P)\| \leq |\lambda| + \|i_n(P) - 4h(P)T\| < \eta + \frac{7}{2} - \frac{\eta}{2} < \frac{7}{2}.$$

Outside $\text{supp } h$, the center vanishes on primitive quotients over $\text{Prim}(I)$, so the distance is at most $\|a\| = 1$; at the quotient character the distance is $|\lambda| \leq 1$. Hence a is in one of the compact-bump balls.

It remains to handle the case in which $|\lambda| \geq \eta$ and the chosen chart has infinite-dimensional fibre. Choose a nonempty open set

$$W = \{P \in \text{Prim}(I) : \|\lambda I + i(P)\| > 3/4\};$$

this is again nonempty because $|\lambda| \geq \eta$ and $\|a\| = 1$. Choose an infinite-dimensional chart $O_n \subset W$ if possible. If every chart inside W has finite-dimensional fibre, the previous paragraph applies, so we may assume H_n is infinite dimensional. Pick $P_0 \in O_n$ and put $k_0 = i_n(P_0)$.

Choose $\mu \in \Lambda$ with $|\mu - \lambda| < \varepsilon$. Then $|\mu| \geq \eta - \varepsilon$. Choose $\omega \in \Omega$ such that

$$|M\omega + \mu| \geq M + |\mu| - \varepsilon.$$

Choose $T \in \mathcal{F}_n$ with $\|T - k_0\| < \varepsilon$. Since T has finite support relative to (e_m) , choose $Q = Q_m \in \mathcal{Q}_n$ with

$$TQ = QT = 0.$$

By norm-continuity of i_n , choose a nonempty open neighbourhood $N \subset O_n$ of P_0 such that

$$\|i_n(P) - k_0\| < \varepsilon \quad (P \in N).$$

Choose $h \in \mathcal{B}_n$ with $\text{supp } h \subset N$, and put

$$c = c_1(n, h, T, Q, \mu, \omega).$$

Since $\|h\|_\infty = 1$, there is $P_1 \in \text{supp } h$ with $h(P_1) = 1$. At that point, the local value of c is

$$\mu I + T + M\omega Q.$$

Because $TQ = QT = 0$, the one-dimensional range of Q reduces this operator and carries the scalar $M\omega + \mu$. Therefore

$$\|c\| \geq |M\omega + \mu| \geq M + |\mu| - \varepsilon \geq M + \eta - 2\varepsilon.$$

We estimate $\|a - c\|$. First let $P \in \text{supp } h$ and set $s = h(P)$. Write

$$e(P) = i_n(P) - T.$$

Then $\|e(P)\| < 2\varepsilon$, and the local difference equals

$$(\lambda - \mu)I + (1 - s)T - sM\omega Q + e(P).$$

The part without $e(P)$ is block diagonal for

$$H_n = QH_n \oplus (1 - Q)H_n,$$

because $TQ = QT = 0$. On QH_n its norm is at most $sM + \varepsilon$, and on $(1 - Q)H_n$ its norm is at most

$$\varepsilon + (1 - s)\|T\| \leq \varepsilon + \|k_0\| + \varepsilon \leq 2 + 2\varepsilon.$$

Thus, since $M = 4$,

$$\|a(P) - c(P)\| \leq M + \varepsilon + 2\varepsilon = M + 3\varepsilon.$$

If $P \in \text{Prim}(I) \setminus \text{supp } h$, then the compact part of c vanishes at P , and

$$\|a(P) - c(P)\| = \|(\lambda - \mu)I + i(P)\| \leq \varepsilon + \|i(P)\| \leq \varepsilon + \|\lambda I + i(P)\| + |\lambda| \leq 2 + \varepsilon < M + 3\varepsilon.$$

At the quotient character of the unitization, the distance is $|\lambda - \mu| < \varepsilon$. Taking the supremum over primitive quotients of A gives

$$\|a - c\| < M + 3\varepsilon.$$

Thus a is covered by one of the scalar-ballast balls.

We have covered S_A by countably many balls. The compact-bump balls have centers of norm 4 and radius $7/2$, hence avoid $B(0, 1/2)$. The scalar-ballast balls have centers at distance at least $\eta - 5\varepsilon > 0$ beyond their radius. Therefore the whole cover is uniformly off the origin, and A has UBCP. $\square$

Scope and Limitations

This is a genuine extension of the continuous-trace separable-fibre theorem, but it is still only a partial result for the source question. It handles one scalar multiplier direction over a positive continuous-trace ideal. It does not yet prove permanence for arbitrary finite-dimensional quotients

$$0 \rightarrow I \rightarrow A \rightarrow F \rightarrow 0$$

or for general type-I/Fell algebras, where matrix-valued multiplier directions and local compression continuity have to be controlled.

The proof suggests the next attack: replace the scalar spike by a finite-dimensional matrix-block spike. If that works, finite-dimensional extensions of the positive continuous-trace class should follow.

Novelty and Search Bounds

The run indexes were searched for arXiv:2311.14257, unitization, finite-dimensional extension, continuous trace, compact operators, scalar multiplier, BCP, and UBCP. Web searches on June 17, 2026 for BCP permanence, finite-codimensional BCP, unitizations, and Fell/type-I local continuous-trace structure found the source paper, the vector-valued $C_0(K, X)$ stability paper arXiv:2111.04921, Calkin/operator-space BCP papers arXiv:2412.07137 and arXiv:2507.02261, and Fell-algebra structure literature arXiv:1207.2540, but no exact finite-dimensional extension or unitized continuous-trace BCP theorem.

Human Review Focus

The main proof points to check are:

- the scalar-ballast estimate in the $|\lambda| \geq 1/16$ case;
- the choice of a finite-support T and a rank-one Q with $TQ = QT = 0$;
- the use of the quotient character of the unitization in the norm estimate;
- the finite-dimensional fibre branch, where the scalar direction is compact and the old continuous-trace proof applies.

If these checks pass, this packet should be treated as a new positive partial result and as the template for the next finite-dimensional quotient push.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Run `fa_banach_001`, lane 19, *BCP for Continuous-Trace C^* -Algebras with Separable Fibres*, solution packet `2311.14257_continuous_trace_separable_fiber_bcp`, 2026.
- [3] Run `fa_banach_001`, lane 19, *A Primitive-Ideal Π -Base Obstruction for C^* -Algebras with BCP*, solution packet `2311.14257_primitive_pi_base_obstruction`, 2026.
- [4] J. Dixmier, *C^* -Algebras*, North-Holland, 1977.
- [5] I. Raeburn and D. P. Williams, *Morita Equivalence and Continuous-Trace C^* -Algebras*, American Mathematical Society, 1998.